

frontend

What this is

Akara is a responsive website (not a native app) that turns NCERT chapters (classes 6 – 12, STEM) into short multilingual animated explainer videos, then closes the loop with a live spoken "explain it back" check (the Feynman technique). It's pure B2C - no school or teacher accounts, no institutional layer. The two core mechanics are:

1. **Video generation** - an NCERT-grounded, citation-backed, multilingual explainer per concept.
2. **The Feynman loop** - the student explains the concept back by voice (or text), and the system checks that explanation against a curated bank of known misconceptions for that concept, not just generic similarity.

Everything else in the product (library, progress, practice formats) is either a cheap byproduct of generating #1, or a UI layer on top of the mastery data produced by #2.

Platform constraints (design for these, don't treat them as edge cases)

- **Website first, PWA-installable.** No Play Store dependency, no APK storage cost, works on any browser. Should be installable to the home screen.
- **Offline chapter packs.** A chapter's videos should be cacheable via service worker once downloaded (e.g. over school Wi-Fi), and usable with no connection after that.
- **Data-saver default.** On a metered/slow connection, default to Listen mode (audio) or lower-bitrate video rather than auto-playing full video. One visible toggle to turn this off.
- **Native-script text input everywhere.** Don't assume English typing.
- **Minimal PII.** Sign-in is Google/Gmail only - name, email, photo. No extra personal data collection. (Most users are minors; this is deliberate.)

- **No social/competitive UI anywhere.** No leaderboards, no rankings, no "vs. classmates." Progress is personal only.

Site map / routes

Route	Purpose
/	Logged-out landing page, "Continue with Google"
/onboarding	One-time: choose class, choose default language
/home	Subject picker, adapts to class (see below)
/home/:subject	Chapter list
/home/:subject/:chapter	Constellation - one node per concept
/learn/:conceptId	Video screen
/explain/:conceptId	Feynman voice/text check
/practice/:conceptId	Format picker: listen / concept card / mind map / quiz
/library	Full multilingual repository - browse/search everything
/progress	Personal mastery dashboard
/profile	Account, language, interests, data settings

Persistent tab bar (bottom on mobile, top on desktop) once past onboarding: **Home · Library · Progress · Profile**. Four tabs, deliberately no more.

Onboarding

1. Sign in with Google (no password flow to build).
2. Choose class (6, 7, 8, 9, 10, 11, 12).
3. Choose default language.

Both choices are editable later from Profile.

Home tab

Subject picker - the subject set depends on class:

- Class 6–10: Science, Maths

- Class 11–12: Physics, Chemistry, Biology, Maths (NCERT splits Science into separate subjects at this level - don't hardcode a single "Science" tile across all classes)

Chapter list - a simple vertical list, not a second constellation. Each row: chapter name + a small ring/progress indicator (concepts mastered / total). Keep this screen cheap to render - the constellation is the one big spatial moment in the app, and putting a second gamified visual here dilutes it and costs render time on low-end devices for no benefit.

Constellation - one node per concept in the chapter. Node states:

State	Meaning
locked	Prerequisite not yet met
available	Unlocked, not yet mastered
mastered	Video watched to completion and Feynman check cleared the mastery threshold
requested	Not yet generated in the student's language, request submitted

Tapping a node does **not** navigate to a new page - it opens a bottom sheet (mobile) / popover (desktop) with three buttons: **Learn, Explain, Practice**. Keep the constellation always one tap away; this matters once "route back to a prerequisite" (see Progress, below) is implemented.

When a node flips to `mastered`, animate a scroll/pan to the next unlocked node. Small easing motion, not a celebration overlay - keep it understated.

Learn screen (`/learn/:conceptId`)

- Video player, clean, no extra chrome.
- A citation line under the player naming the source NCERT chapter/section - this is a trust signal, don't cut it for space.
- Language toggle.
- One clear CTA: "Explain it back" → goes to `/explain/:conceptId`.

Loading states to design for (this reflects how the backend actually serves video, relevant for what message/spinner to show):

- **Instant** - exact concept + language already exists in the library. No generation happens; this should feel like a normal video load, not an AI-generation wait state.
- **Short wait ("finishing the dub")** - the concept exists in another language already; only audio is being re-synthesized. Give this its own lighter loading copy, it should be fast.
- **Longer wait ("generating this for the first time")** - genuine cache miss, full pipeline runs. This is the only case that needs a real progress indicator / can't-be-instant messaging.

Explain screen (`/explain/:conceptId`)

- Mic-first UI: a large press-to-speak button, waveform/listening indicator.
- A mentor persona bubble shows the prompt or follow-up question, in the student's language. Two response modes to design for: a mastery confirmation, or a targeted follow-up tied to a specific known misconception (sometimes phrased as the mentor stating something wrong and asking the student to catch it - same UI, just different bubble content).
- **A "type instead" option must sit next to the mic, not buried in settings.** A shared-room household isn't always a place to speak answers aloud on command; text input in the student's own script needs to be equally first-class.

Practice screen (`/practice/:conceptId`)

A picker, four cards/icons, each its own lightweight destination:

- **Listen** - the video's audio track alone, no video.
- **Concept card** - a short text summary generated from the same script.
- **Mind map** - a static visual of how this concept relates to others, generated from the same scene-graph data used to build the video (not a new content type to author).
- **Quick quiz** - a few auto-generated questions from the same script.

None of these are new subsystems - they're different renders of data already produced when the video was generated. Build them as lightweight, not as four separate content pipelines.

Progress logic (root-cause remediation)

If a student fails the same misconception on a concept more than once, don't just re-ask a harder question about the same node - route them to the actual prerequisite node in the constellation graph instead (existing library content, no new generation). The constellation is a real prerequisite graph, not just a decorative map; the Explain flow should be able to read it and redirect.

Library tab (/library)

The searchable, browsable face of the entire multilingual repository - not just an internal cache. Structure:

- Grouped by subject.
- Each concept/node is searchable by name.
- Two display modes the user can toggle: **my progress** (colored by personal mastery state) and **language coverage** (colored by how many languages exist for that concept - this is real data, not decorative).
- A live counter of total videos generated across all languages.
- Tapping a concept opens the same three-way action sheet (Learn / Explain / Practice) as Home.

This screen is allowed to be visually rich - it's the exploration/discovery surface, not a daily-use screen, so the "wow" budget is spent here rather than on the chapter list.

Progress tab (/progress)

A personal-only dashboard, not a leaderboard:

- A compact visual of only the concepts the student has touched (mastered / available / needs-revisit), organized by subject.

- Three summary numbers: concepts mastered, concepts to revisit, active days this week.
- An actionable "worth another look" list - concepts flagged by the misconception log - each with a direct link back into that concept's constellation node. This list is driven by real diagnostic data, not a generic "you haven't finished this" nudge.

Profile tab (</profile>)

- Account block: name/photo/email from Google, current class.
- Default language, with a "change" action.
- Interest tags (e.g. cricket, farming, Bollywood, gaming) - used to personalize the examples inside generated explanations. Editable chip list.
- **Request a language** - a text field plus a live note like "N students requested [language] this month" - this isn't decorative copy, it's the real demand signal that should feed the batch-generation queue for what gets rendered next.
- Data saver toggle (audio-preferred on mobile data).
- Downloaded/offline chapters, with a manage action.
- A one-line, expandable disclosure: "How Akara checks your understanding" - a plain-language explanation of the misconception-bank diagnosis. Keep this visible in normal settings, not buried in a ToS page - it's a transparency signal worth surfacing.
- Sign out.

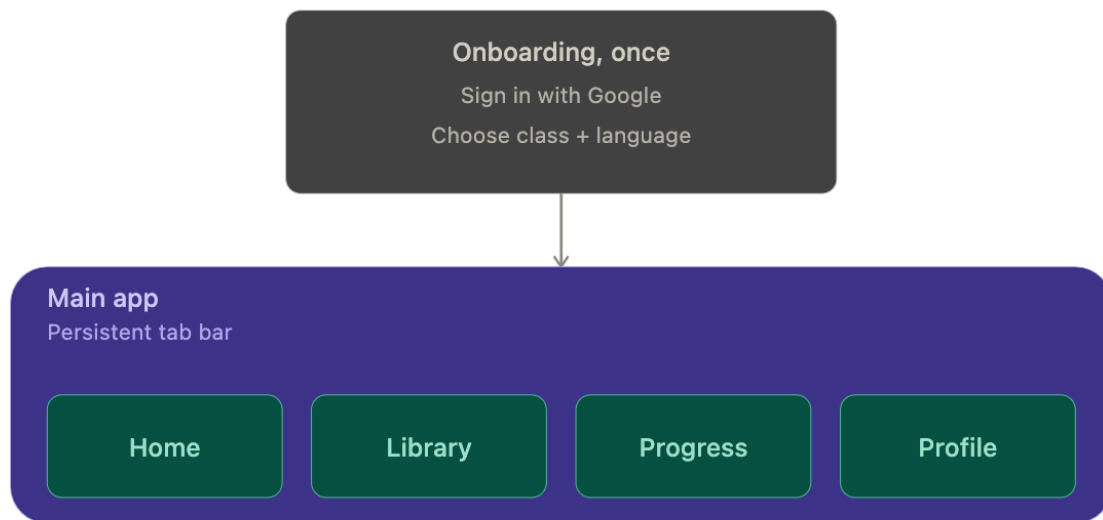
Profile is deliberately the plainest screen in the app - it doesn't need visual richness, it needs to be fast and trustworthy.

Explicitly out of scope

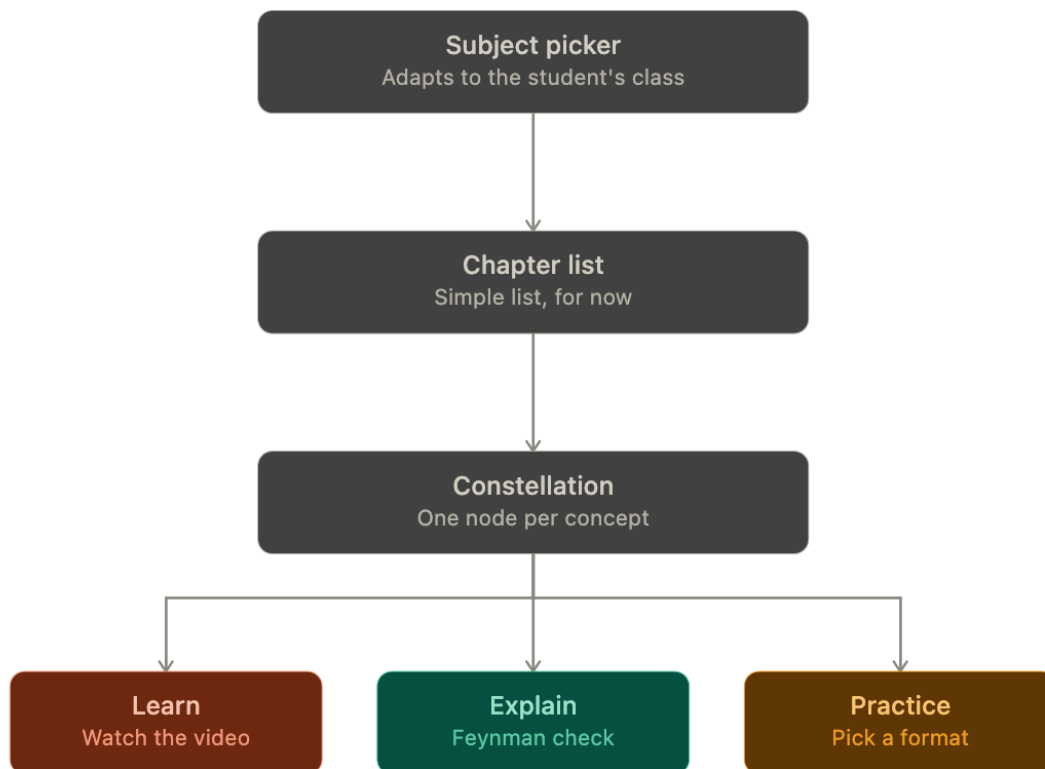
- No teacher, facilitator, or school-admin views of any kind.
- No classroom/cohort features, no social comparison, no leaderboards.
- No distribution or deployment layer for schools - this is a self-serve, individual-student product.

Screens/visuals referenced in this brief

Mockups for the constellation, video screen, Feynman screen, library ("knowledge galaxy"), progress ("your sky"), and profile screens were prototyped earlier in this design session and will be attached separately - this document describes their structure and behavior, not their exact pixel layout.



Onboarding Flow



home page

Search a chapter or concept

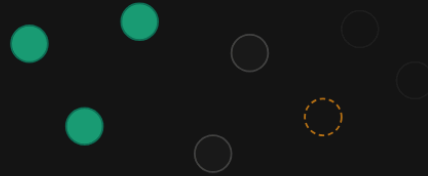
My progress

Language
coverage

Physics



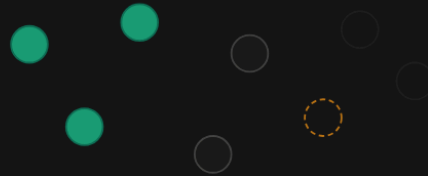
Chemistry



Biology

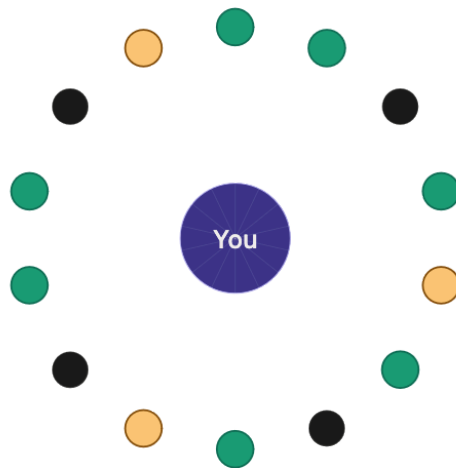


Maths



8,540 videos generated, across 12 languages, and

library



Concepts mastered

7

To revisit

3

Active this week

4 days

Worth another look

Acids and bases

Revisit

Magnetism

Revisit

Heredity

Revisit

progress



Riya
riya@gmail.com · class
10

Default language
हिंदी

Change

Interests, for personalized examples

Cricket

Farming

Add

Don't see your language?

Request a language

Request

127 students requested Odia for this class this month, queued for the next generation batch

Data saver

Prefer audio over video on mobile
data

Downloaded chapters

2 chapters, 340 MB

Manage

How Akara checks your understanding



Sign out

settings